Modelling the Perioperative Care Pathway

STRESS reporting of the Perioperative Pathway Explorer

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1. Introduction

1.1 Background

Perioperative care - the care provided before, during, and after surgery - is a complex and essential part of delivering high-quality elective care. While there has been significant progress through developments such as elective hubs, new technologies, and pathway redesign, the overall effectiveness of these initiatives depends on how well perioperative care supports patients across the whole pathway. This includes early identification, optimisation, and coordinated care to improve outcomes and support timely recovery.

A key challenge is that the impact of perioperative care is often spread across the system and not always easy to measure. While some outcomes, such as reduced length of stay, are routinely captured, others - such as improved long-term health, reduced complications, and better use of capacity - are less visible. Early screening and assessment are particularly important, as they create opportunities to identify risk, intervene sooner, and improve patient readiness for surgery, but their downstream benefits are often under-recognised.

Limited capacity, inconsistent early screening and delays in discharges in perioperative care are resulting in an accumulation of patients on waiting lists. Longer waits are associated with poor health outcomes, rising risk of complications and prolonged hospital stays.

Within the NHS, the perioperative care pathway is a complex, interconnected system, spanning multiple disciplines and organisations. Modelling the whole pathway can improve the understanding of how changes in one part of the system can have ripple effects across the entire system.

1.2 Project overview

This project takes a participative approach to support the development of a decision-support simulator (Perioperative Pathway Explorer) which allows users to visualise the effect of interventions across the whole pathway, and test scenarios in a risk-free environment. The specific focus of this project is on elective hip and knee surgical pathways.

Built using a system dynamics methodology, this high-level tool brings together evidence-based information with practice- and expert-informed insights to support healthcare staff, including service planners and commissioners, in understanding the benefits of early screening and other interventions (including health improvement activities).

It has been developed with NHS medical, nursing and service commissioning experts and includes input from the Centre for Perioperative Care, the Royal College of Anaesthetists, the Royal College of Surgeons, GIRFT and leads from Integrated Care Boards.

2. Documenting the SD model

2.1 Objectives

2.1.1 Purpose of model

The purpose of the model is to study the effects of interventions at different points along the perioperative care pathway for elective hip and knee replacements.

2.1.2 Model Output

The outputs of the model are the size of waiting lists at different points along the pathway, and by whether they have been on an early screening pathway.

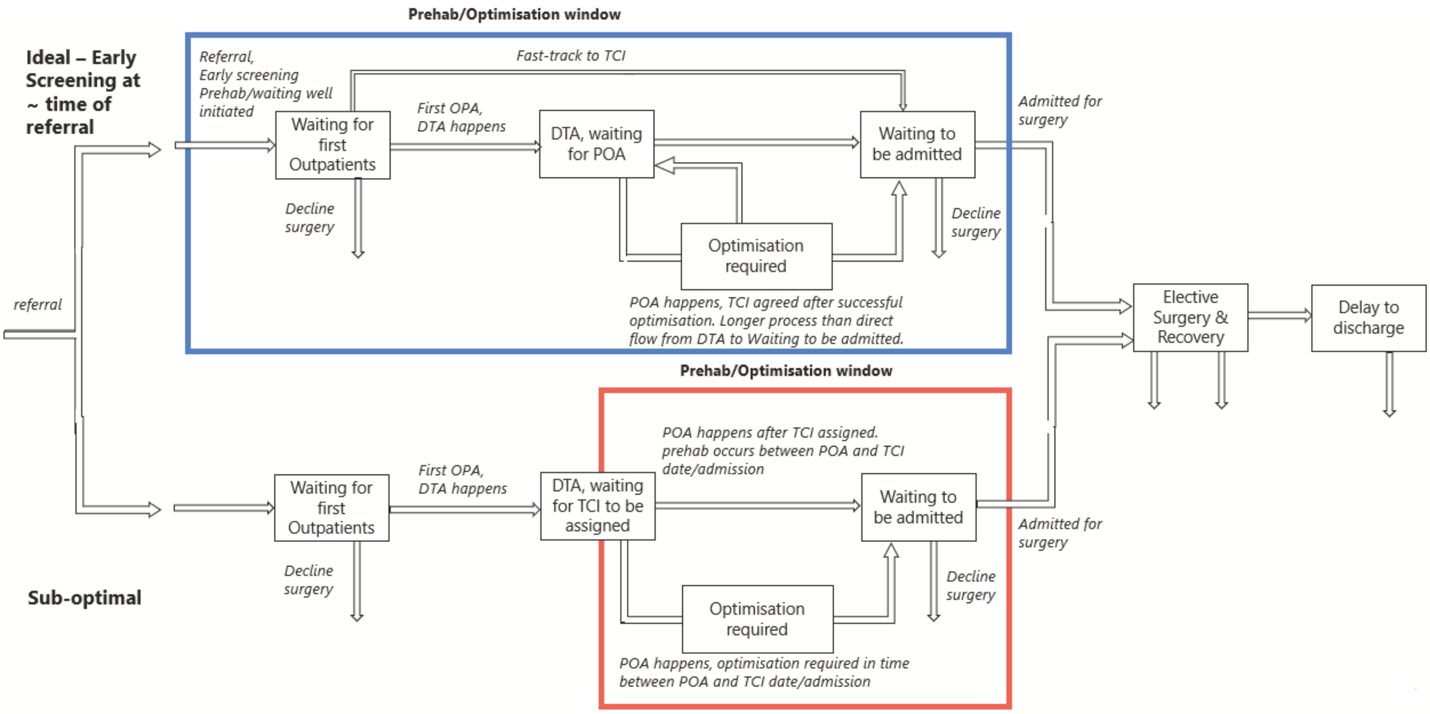
2.1.3 Experimentation Aims

Through workshops, three different interventions were identified, as detailed in the table below. The design of the interface allows for these interventions to be applied for the whole simulation period, or adjusted at specific points in the future using the ballistic mode.

Intervention	Quantify in the model	Notes
Primary prevention strategies	Apply an adjustment to the annual referral rate per head of population	A 'catch-all' that could incorporate multiple strategies
Increased early screening & assessment	Increase proportion of new referrals routed to early screening & assessment	Improved time for medical optimisation, reduces likelihood of last-minute surgical cancellations and length of stay
Increase early discharge planning	Apply a percentage to the number of daily surgical admissions	Reduces the proportion of recovered patients that have a delayed discharge

2.2 Logic

2.2.1 Base model overview diagram



2.2.2 Base model logic

A proportion of the adult population are referred to Orthopaedics & Trauma each year, of which a percentage are specifically for issues with hips and knees. Some of these patients go through an early screening programme and associated pathway; the remainder follow the 'traditional' pathway.

For both pathways the steps are: referral -> first outpatient appointment (OPA) -> Decision to Treat (DTA) -> Preoperative assessment (POA) (during which some patients are identified as requiring medical optimisation) -> Receive 'To Come In' date (TCI) -> Admission for surgery & recovery -> discharge, with some patients having a delay in their discharge.

Patients may choose to leave the pathway after the first outpatient appointment or whilst waiting to be admitted (conservative care, decline surgical options or seek treatment privately).

The difference in the two pathways occurs between first outpatient appointment and waiting to be admitted. In the early screening pathway, some patients are identified as having low complexity and can be fast-tracked to wait to be admitted. For patients that require medical optimisation, there is the opportunity for those in the early screening to extend their time being optimised, whereas those on the traditional pathway automatically wait to be admitted after the optimisation period. This leads to a proportion of these patients being unfit for surgery and having their surgery cancelled within 24 hours of surgery date. These patients re-join the stock waiting to be admitted.

After surgery, most patients have a timely discharge but some are delayed.

2.2.3 Intervention logic

There are three intervention points:

- reduce referrals coming into the system
- increase the proportion of new referrals that undergo early screening and assessment
- increase the proportion of admitted patients who undergo early discharge planning

2.2.4 Algorithms

NA

2.2.5 Components

2.2.5.1 Stocks

Units: People. Initial values are set from the calcs submodel at model start (calcs.init_* variables).

Stock	Initial value eqn	Inflows	Outflows	Role
i wait opa	calcs.init_i_wait_opa	i_referrals	i_do_not_proceed_surgery; i_opa_to_poa; i_fasttrack_to_admit	Ideal-pathway (early-screened and assessed) patients waiting for their first outpatient appointment (OPA), following referral.
s wait opa	calcs.init_s_wait_opa	s_referrals	s_do_not_proceed_surgery; s_opa_to_tci	Sub-optimal-pathway (not early-screened) patients waiting for first OPA.
i wait poa	calcs.init_i_wait_poa	i_opt_to_poa; i_opa_to_poa	i_poa_to_ready; i_poa_to_opt	i-pathway patients waiting for pre-operative assessment (POA), having come directly from OPA or been referred back from optimisation (OPT).
i wait opt	calcs.init_i_wait_opt	i_poa_to_opt	i_opt_to_poa; i_opt_to_ready	i-pathway patients in the "optimisation" (OPT) sub-pathway, addressing a fitness issue identified at POA before surgery.
i wait with tci	calcs.init_i_wait_admit	i_poa_to_ready; i_opt_to_ready; i_fasttrack_to_admit	i_admit; i_preadmit_dropout	i-pathway patients cleared and waiting for admission ("TCI" = To Come In / ready pool).
s tci wait poa	calcs.init_s_wait_poa	s_opa_to_tci	s_poa_to_opt; s_poa_to_ready	s-pathway patients listed for surgery (TCI) and waiting for POA.
s wait opt	calcs.init_s_wait_opt	s_poa_to_opt	s_opt_to_ready	s-pathway patients in the OPT sub-pathway.

Stock	Initial value eqn	Inflows	Outflows	Role
surgery and recovery	calcs.init_treat	s_admit; i_admit	same_day_cancellation_need_reopt; timely_discharge; medically_fit_but_delayed	Patients currently admitted — in theatre or in post-operative recovery (both pathways combined).
s wait post poa	calcs.init_s_wait_admit	s_poa_to_ready; s_opt_to_ready; s_reopt	s_admit; s_preadmit_dropout	s-pathway patients cleared at POA (or re-listed after a same-day cancellation) and waiting for admission.
delayed discharge	calcs.init_delay	medically_fit_but_delayed	supported_discharge	Patients medically fit for discharge but remaining in a bed pending supported/social discharge arrangements.

2.2.5.2 *Flows*

Units: People/Days. "Type" indicates whether the flow is a model boundary (source/sink) or internal to the pathway; see §2.2.5.5 for the consolidated source/sink list and two important caveats (marked ** below).

Flow	Type / connection	Equation	Role
i admit	Internal — i_wait_with_tci → surgery_and_recovery	IF surg_demand>0 THEN MIN(i_wait_with_tci, cap_theatre) * (i_wait_with_tci / surg_demand) ELSE 0	Admits i-pathway patients to surgery; shares theatre capacity with s-pathway pro-rata to each pathway's share of total ready demand.
referrals	Unconnected* — —	referrals_per_day * p_refs_to_hips_knees	Total daily referrals filtered to the hips & knees pathway. *Not itself attached to a stock — feeds the i/s referral split below.
i referrals	Source — (external) → i_wait_opa	referrals * early_screening	New referrals routed to the i-pathway (share set by early_screening).

Flow	Type / connection	Equation	Role
s referrals	Source — (external) → s_wait_opa	referrals * (1 – early_screening)	New referrals routed to the s-pathway (remaining share).
i poa to ready	Internal — i_wait_poa → i_wait_with_tci	IF poa_demand>0 THEN MIN(i_wait_poa, cap_poa)*(i_wait_poa/poa_demand)*(1−p_i_unfit) ELSE 0	i-pathway patients cleared as fit at POA, moving to the ready-for-admission pool.
i poa to opt	Internal — i_wait_poa → i_wait_opt	IF poa_demand>0 THEN MIN(i_wait_poa, cap_poa)*(i_wait_poa/poa_demand)*(p_i_unfit) ELSE 0	i-pathway patients found unfit at POA, routed to OPT.
i opt to ready	Internal — i_wait_opt → i_wait_with_tci	(i_wait_opt/opt_time) * (1 – p_i_re_poa)	i-pathway patients completing OPT and becoming ready for admission (time-based delay via opt_time).
s opa to tci	Internal — s_wait_opa → s_tci_wait_poa	IF opa_demand>0 THEN MIN(s_wait_opa, cap_opa)*(s_wait_opa/opa_demand)*(1−p_non_surgery) ELSE 0	s-pathway patients seen at OPA and listed for surgery, moving into the TCI/POA wait.
s poa to opt	Internal — s_tci_wait_poa → s_wait_opt	IF poa_demand>0 THEN MIN(s_tci_wait_poa, cap_poa)*(s_tci_wait_poa/poa_demand)*(p_s_unfit) ELSE 0	s-pathway patients found unfit at POA, routed to OPT.
s opt to ready	Internal — s_wait_opt → s_wait_post_poa	s_wait_opt / opt_time	s-pathway patients completing OPT.
s poa to ready	Internal — s_tci_wait_poa → s_wait_post_poa	IF poa_demand>0 THEN MIN(s_tci_wait_poa, cap_poa)*(s_tci_wait_poa/poa_demand)*(1−p_s_unfit) ELSE 0	s-pathway patients cleared as fit directly at POA.
s admit	Internal — s_wait_post_poa → surgery_and_recovery	IF surg_demand>0 THEN MIN(s_wait_post_poa, cap_theatre)*(s_wait_post_poa/surg_demand) ELSE 0	Admits s-pathway patients to surgery, sharing theatre capacity with i-pathway.

Flow	Type / connection	Equation	Role
i opt to poa	Internal (rework) — i_wait_opt → i_wait_poa	(i_wait_opt/opt_time) * p_i_re_poa	i-pathway patients requiring re-assessment at POA after OPT.
s do not proceed surgery	Sink — s_wait_opa → (external)	IF opa_demand>0 THEN MIN(s_wait_opa, cap_opa)*(s_wait_opa/opa_demand)*p_non_surgery ELSE 0	s-pathway patients seen at OPA who do not proceed to surgery; leave the tracked pathway.
i do not proceed surgery	Sink — i_wait_opa → (external)	IF opa_demand>0 THEN MIN(i_wait_opa, cap_opa)*(i_wait_opa/opa_demand)*p_non_surgery ELSE 0	i-pathway equivalent of the above.
medically fit but delayed	Internal — surgery_and_recovery → delayed_discharge	medically_fit * delay_fraction	Patients becoming medically fit whose discharge is delayed (fraction delay_fraction).
timely discharge	Sink — surgery_and_recovery → (external)	medically_fit * (1 − delay_fraction)	Patients discharged without delay; leaves the tracked system (model output).
supported discharge	Sink — delayed_discharge → (external)	delayed_discharge / delay_los	Delayed patients eventually discharged with support, at a rate set by average delay length (delay_los).
same day cancellation need reopt	Sink** (feeds s_reopt) — surgery_and_recovery → (s_reopt)	s_admit * s_cancel_reopt_rate	Same-day cancellations needing re-optimisation. **Not a true model boundary — the volume re-enters via s_reopt (see §2.5.5).
i fasttrack to admit	Internal — i_wait_opa → i_wait_with_tci	IF opa_demand>0 THEN MIN(i_wait_opa, cap_opa)*(i_wait_opa/opa_demand)*p_fasttrack*(1−p_non_surgery) ELSE 0	"Fast-track" i-pathway patients moving directly from OPA to the ready-for-admission pool, bypassing POA/OPT.

Flow	Type / connection	Equation	Role
i opa to poa	Internal — i_wait_opa → i_wait_poa	IF opa_demand>0 THEN MIN(i_wait_opa, cap_opa)*(i_wait_opa/opa_demand)*(1−p_fasttrack)*(1−p_non_surgery) ELSE 0	Non-fast-track i-pathway patients moving from OPA into POA.
s reopt	Source** (rework) — (same_day_cancellation_need_reopt) → s_wait_post_poa	same_day_cancellation_need_reopt	Returns same-day-cancelled s-pathway patients back into the ready-for-admission pool. **A rework loop, not a true external source — see §2.2.5.5.
s preadmit dropout	Sink — s_wait_post_poa → (external)	s_wait_post_poa * dropout_rate_alt	s-pathway patients who drop out while waiting for admission.
i preadmit dropout	Sink — i_wait_with_tci → (external)	dropout_rate_alt * i_wait_with_tci	i-pathway equivalent of the above.

2.2.5.3 Constants/Converters/auxiliary variables

Operational parameters (structurally determine flow rates)

29 converters that are referenced, directly or transitively, by at least one flow or stock equation. Five of these (marked †) are placeholder interface variables in the main model whose live value is actually supplied by the calcs submodel via a module connection — their equations here show the calcs-side calculation instead of the main-model placeholder text.

Converter	Units	Value / equation	Role
base delay los	Days	1	Baseline (unimproved) length of stay attributable to discharge delay, before any discharge-support effect is applied.

Converter	Units	Value / equation	Role
cap opa †	People	Module input from calcs (see below)	Daily capacity for outpatient (OPA) appointments; value supplied via module connection from the calcs submodel.
cap poa	People	10000	Daily capacity for pre-operative assessment; large constant, effectively unconstrained by default.
cap theatre †	People	Module input from calcs (see below)	Daily theatre (surgical) capacity shared by both pathways; supplied via module connection from calcs.
delay fraction	dmnl	0.03 + (.12*(1−early_screening_1)*(1−(discharge_support_effect...)))	Proportion of medically-fit patients whose discharge is delayed; sensitive to early-screening rate and discharge-support effect.
delay los	Days	(improved_delay_los*discharge_support_effect/100) + (base_delay_los...)	Average discharge-delay length, blending base and "improved" (supported) durations.
discharge support effect	dmnl	1	Scaling factor for the effectiveness of discharge-support interventions (1 = baseline); a user intervention point.
dropout rate alt	dmnl	0.002 (standalone default) — supplied by calcs.calib_dropout_rate when linked (see below)	Daily probability a patient waiting for admission drops out / is removed from the pathway.
early screening †	dmnl	Module input from calcs (see below)	Proportion of new referrals routed to the i (fast-track) pathway; supplied via module connection from calcs.
i early screening los	Days	0.5	Reduction in length of stay associated with early-screened (i-pathway) patients.

Converter	Units	Value / equation	Role
improved delay los	Days	0.4	Achievable discharge-delay length under effective discharge support.
medically fit	People/Days	surgery_and_recovery / treatment_los	Rate at which admitted patients become medically fit for discharge.
new treatment los	Days	s_treatment_los – screening_reduction_in_los	Adjusted treatment length of stay after applying the early-screening LOS reduction.
opa demand	People	i_wait_opa + s_wait_opa	Total patients (both pathways) competing for OPA capacity; used to pro-rate OPA-stage flows.
opt time	Days	42	Average duration patients spend in the OPT sub-pathway.
p fasttrack	dmnl	0.2	Proportion of i-pathway OPA patients routed directly to the ready pool (fast-track).
p i re poa	dmnl	0.2	Proportion of i-pathway OPT patients requiring re-assessment at POA.
p i unfit	dmnl	0.25	Proportion of i-pathway patients found unfit at POA.
p non surgery	dmnl	0.555082247666873	Proportion of OPA patients (either pathway) who do not proceed to surgery.
p refs to hips knees	dmnl	0.189	Proportion of all O&T referrals relevant to the hips & knees specialty.
p s unfit	dmnl	0.4	Proportion of s-pathway patients found unfit at POA.
poa demand	People	i_wait_poa + s_tci_wait_poa	Total patients (both pathways) competing for POA capacity.

Converter	Units	Value / equation	Role
referrals per day †	People/Days	Module input from calcs (see below)	Total daily referral rate across all specialties; supplied via module connection from calcs.
s cancel reopt rate	dmnl	0.1	Proportion of s-pathway admissions cancelled same-day, requiring re-optimisation.
s treatment los	Days	1.5	Baseline s-pathway treatment length of stay before screening adjustment.
screening reduction in los	Days	s_treatment_los − i_early_screening_los	LOS reduction attributable to early screening.
surg demand	People	i_wait_with_tci + s_wait_post_poa	Total patients (both pathways) ready and competing for theatre/admission capacity.
treatment los	Days	new_treatment_los*early_screening_1 + ((1−early_screening_1)*s_treatment_los)	Blended treatment LOS used in the medically-fit rate, weighted by early-screening proportion.

† Module-imported values (calculated in the calcs submodel)

Main-model interface variable	Module connection	Equation (in calcs)
cap opa	cap_opa ← calcs.cap_opa	IF TIME<=lead_time THEN calib_cap_opa ELSE new_opa_capacity
cap theatre	cap_theatre ← calcs.theatre_cases_pd	IF TIME>lead_time THEN cap_nhs_theatre/(1−p_theatre_private) ELSE cap_nhs_theatre/(1−p_theatre_private_init)

Main-model interface variable	Module connection	Equation (in calcs)
early screening / early screening 1	both ← calcs.early_screening	IF TIME>lead_time THEN pc_early_screening/100 ELSE 0.4
referrals per day	referrals_per_day ← calcs.referrals_per_day	referrals_per_year / 365
dropout rate alt	sourced from calcs.calib_dropout_rate at initialisation	0.000031235060138762 (calibrated constant)

Output / reporting converters (excluded from the operational core)

Provided for reference only. These 27 converters in the main model are computed from stocks/flows for monitoring and reporting purposes but are not themselves referenced by any flow or stock equation, so they do not affect how the model runs:

activity per day (admissions less late cancel) · admit · ave time dta to treat (weeks) · ave time ref to opa (weeks) · ave time rtt (weeks) · cases per 4-hr session · clock stops · clock stops dta no treat · clock stops no treat no dta · clock stops no treatment all · dropout rate · i clock stops no treatment · i in pathway · i waiting with dta · in hospital · lead time · new dta · opt demand · pc early screening · preadm drops · s clock stops no treatment · s in pathway · s waiting with dta · total in pathway · total waiting · treated · waiting with dta

2.2.5.4 Graphical functions

All 8 graphical functions in the model reside in the calcs submodel, where they hold historical/published data used to calibrate model parameters against real-world benchmarks (see module-imported values above, e.g. cap_opa, cap_theatre, referrals_per_day, early_screening). None of these graphical functions is itself passed directly into main — they inform the calcs-side calibration calculations that in turn feed the module-imported converters.

Graphical function (in calcs)	Content	Role
pub rtt o&t all	Published RTT (referral-to-treatment) waiting-time data, orthopaedics & trauma — all pathways.	Calibration target within calcs
pub rtt o&t dta	Published RTT data specific to the decision-to-admit (DTA) stage, orthopaedics & trauma.	Calibration target within calcs
pub rtt wait opa	Published RTT data for waiting times to first outpatient appointment.	Calibration target within calcs
wlmds flow new rtt	NHS Waiting List Minimum Data Set (WLMDS) — flow of new RTT pathway starts.	Calibration target within calcs
wlmds cum new rtt	WLMDS — cumulative new RTT pathway starts.	Calibration target within calcs
wlmds cum stops admit	WLMDS — cumulative pathway stops via admission.	Calibration target within calcs
wlmds cum stops no admit	WLMDS — cumulative pathway stops without admission.	Calibration target within calcs
wlmds cum stops with dta no treat	WLMDS — cumulative stops with a decision-to-admit but no treatment.	Calibration target within calcs

2.2.5.5 Sources and sinks

Sources

Flow	Enters stock	Description
i referrals	i_wait_opa	External population of new i-pathway referrals entering the tracked system.
s referrals	s_wait_opa	External population of new s-pathway referrals entering the tracked system.

Sinks

Flow	Leaves stock	Description
s do not proceed surgery	s_wait_opa	s-pathway patients leaving after OPA without surgery.
i do not proceed surgery	i_wait_opa	i-pathway patients leaving after OPA without surgery.
timely discharge	surgery_and_recovery	Patients discharged without delay.
supported discharge	delayed_discharge	Delayed patients discharged with support.
s preadmit dropout	s_wait_post_poa	s-pathway patients dropping out pre-admission.
i preadmit dropout	i_wait_with_tci	i-pathway patients dropping out pre-admission.

All sink variables are clock stops.

2.3 Data

2.3.1 Data sources

Source name	Date range	Description
Office of National Statistics	2025-2035	Population estimates and projections of ages 18+ for England
RTT Waiting Times	2025	Monthly report of: new referrals (clock starts), clock stops, open pathways, open pathways with DTA
Secondary Uses Service (SUS)	2025	Treated clock stops by procedure and provider type
Waiting List Minimum Dataset (WLMDS)	2025	Patient-level dataset of clock starts, clock stops by waiting list type.
NHSE National Theatres Productivity dashboard	2025	Planned and actual number of 4-hour sessions per week, number of cases per session

2.3.2 Pre-processing

WLMDS: monthly totals of clock starts, clock stops by waiting list type

SUS: derive hip/knee replacements as percentage of O&T completed procedures; apply that % to pathways with DTA (open pathways, admitted clock stops and clock stops with DTA but no treatment).

SUS: for completed hip/knee replacements, derive % split by NHS/independent

RTT: derive pathways waiting for first OPA (based on assumption in 2.3.4)

2.3.3 Input parameters

Main-model input parameters (base case)

15 exogenous constants directly used to run the model. † denotes a parameter that is overridden by a module-imported value from calcs when the model is run linked (as opposed to main being tested standalone) — see §2.2.5.3 for the full module-import mapping.

Parameter	Units	Base case value	Description of use
base delay los	Days	1	Baseline discharge-delay duration before any benefit from a discharge-support intervention is applied.
cap poa	People	10000	Daily cap on pre-operative assessment (POA) capacity; effectively unconstrained in the base case.
discharge support effect	dmnl	1	Baseline (no-change) scaling factor for the effectiveness of discharge-support interventions. Primary candidate for policy/scenario variation — see note below.

Parameter	Units	Base case value	Description of use
dropout rate alt †	dmnl	0.002 standalone default; 0.000031235060138762 when linked (from calcs.calib_dropout_rate)	Daily probability a patient drops out while waiting for admission.
i early screening los	Days	0.5	Length-of-stay reduction associated with the ideal (early-screened) pathway.
improved delay los	Days	0.4	Achievable discharge-delay duration under effective discharge support.
opt time	Days	42	Average duration patients spend in the optimisation (OPT) sub-pathway.
p fasttrack	dmnl	0.2	Proportion of ideal-pathway OPA patients fast-tracked directly to the ready-for-admission pool.
p i re poa	dmnl	0.2	Proportion of ideal-pathway OPT patients requiring re-assessment at POA.
p i unfit	dmnl	0.25	Proportion of ideal-pathway patients found unfit at POA (routed to OPT).
p non surgery	dmnl	0.555082247666873	Proportion of OPA patients (either pathway) who do not proceed to surgery.
p refs to hips knees †	dmnl	0.189	Proportion of total daily referrals relevant to hips & knees; also fed out to calcs (overriding its own internal default) to size OPA-capacity calibration — see below.
p s unfit	dmnl	0.4	Proportion of sub-optimal-pathway patients found unfit at POA.
s cancel reopt rate	dmnl	0.1	Proportion of sub-optimal-pathway admissions cancelled same-day, requiring re-optimisation.
s treatment los	Days	1.5	Baseline sub-optimal-pathway treatment length of stay, before the screening LOS adjustment.

Calcs-submodel input parameters (feeding calibration / module-imported values)

17 exogenous constants within calcs that ultimately determine the module-imported values described in §2.5.3 (cap_opa, cap_theatre, early_screening, referrals_per_day, dropout_rate_alt). These are the base-case data underlying the calibration — i.e. "where the numbers came from" for the imported operational parameters.

Parameter	Units	Base case value	Description of use
4-hr sess per week calib	sessions/week	3542	Calibration-period count of 4-hour theatre sessions per week (used while TIME ≤ lead_time).
new 4-hr sess per week	sessions/week	3542	Post-calibration count of 4-hour theatre sessions per week (used once TIME > lead_time); identical to the calibration value in this file.
cases per sess calib	People/session	1.9947371	Calibration-period theatre throughput per 4-hour session.
cases per 4-hr session	People/session	1.99	Post-calibration theatre throughput per 4-hour session, used once TIME > lead_time.
pc theatre efficiency	dmnl	90.2 (%)	Theatre efficiency percentage applied once TIME > lead_time.
th eff calib	dmnl	0.902	Theatre efficiency fraction applied during the calibration period.
p theatre private init	dmnl	0.33	Initial (calibration-period) proportion of theatre activity assumed private.
percent theatre private	dmnl	33 (%)	Post-calibration proportion of theatre activity assumed private.
p o&t treated hips knees	dmnl	0.28	Proportion of orthopaedics & trauma (O&T) theatre activity attributable to hips & knees.
o&t referral rate p 10th pa	Per Year	320	Annual O&T referral rate per 10,000 population, used to derive the total referral volume.
diversion ref rate per 10th	Per Year	0	Referral rate per 10,000 population diverted away from the pathway; zero in this file (no diversion scheme modelled).
lead time	Days	366	Cutover day separating the calibration period (TIME ≤ lead_time, matched to historical data) from the projected period (TIME > lead_time).
p alt dta pc rtt	dmnl	0.2	Alternative decision-to-admit (DTA) proportion of RTT, used if sw_alt_dta_pc_rtt is switched on.
sw alt dta pc rtt	dmnl (switch)	0	Switch selecting between the published DTA/RTT ratio (0, default) and the alternative fixed ratio p_alt_dta_pc_rtt (1).
pc early screening	dmnl	40 (%)	Post-calibration early-screening percentage, used once TIME > lead_time (overridden by main's identical value when linked — see note).

Parameter	Units	Base case value	Description of use
p refs to hips knees (calcs default)	dmnl	0.3125653	Standalone default only — overridden by main.p_refs_to_hips_knees (0.189) whenever the module is linked; see note below.
calib dropout rate	dmnl	0.0000312	Calibrated dropout-rate value; feeds main.dropout_rate_alt via module connection.

2.3.4 Assumptions

- referral rates are constant throughout the simulation period (unless manually adjusted)
- probabilities are fixed throughout the simulation period
- all RTT without a DTA are waiting for first OPA
- capacity constraints (OPA and theatre) are approximated by flow rates rather than scheduling
- hips & knees theatre allocation is in the same proportion as treated clock stops

2.4 Experimentation

2.4.1 Initialisation

All stocks are initialised with the January 2025 value of their equivalent WLMDS value.

2.4.2 Run length

- Run length: 4015
- DT: 0.125
- Time unit: days

2.4.3 Estimation approach

One iteration of Powell optimisation is used to calibrate:

1. percentage of referrals that are specific to hips & knees
2. percentage that decline surgical pathway after first outpatient appointment
3. drop-out rate whilst waiting to be admitted

2.5 Implementation

2.5.1 Software

Stella Architect, version 4.3.1, build 3962

2.5.2 Random sampling

NA

2.5.3 Model execution

- Integration method: Euler
- Time step/DT: 0.125

2.5.4 System specification

- Windows 10 Pro, version 22H2, OS Build: 19045.5247
- 11th Gen Intel(R) Core(TM) i7-11800H @ 2.30GHz 2.30 GHz, RAM 32.0GB

2.6 Code Access

2.6.1 Computer model sharing statement

The Stella Architect stmx model file is available in a repository on the Strategy Unit's Github account. Ultimate ambition is for the model and interface to be converted to open source, but this will still read in and use the stmx model.